

**Tribhuvan University**

BSc CSIT 7th Semester

**Data Warehousing and Data Mining (CSC410)**

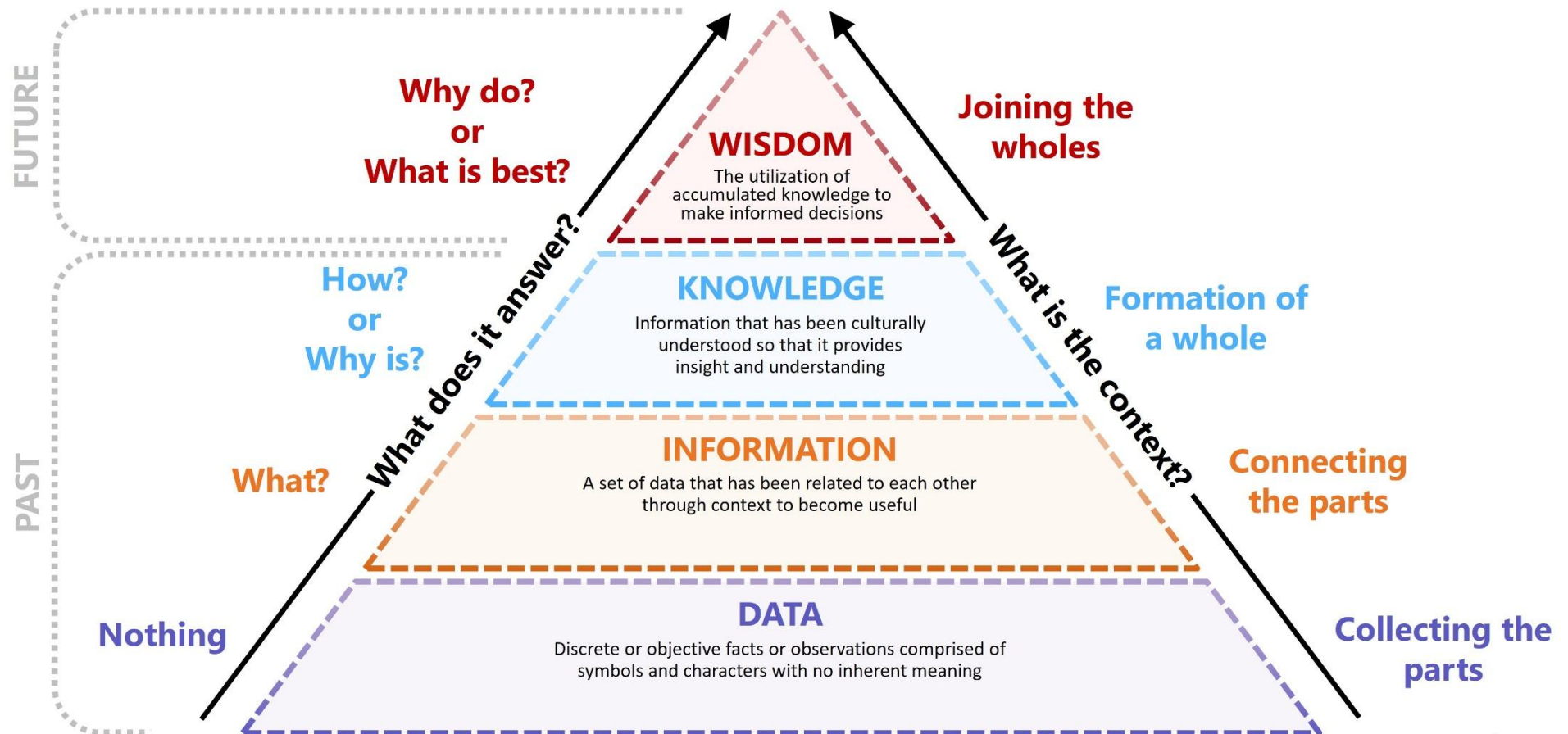
**Unit 1:**

**Introduction to Data Warehousing**

By: Dabbal Singh Mahara

Amrit Campus

# DIKW Pyramid



# DIKW

- **Data**

- Discrete or objective facts or observations comprised of symbols and characters with no inherent meaning.
- Data alone doesn't tell a story. It's the raw material that needs structure before it can be understood.

- **Information**

- A set of data that has been related to each other through context to become useful.
- Information adds context, but it still only describes what happened. It doesn't yet explain why.

- **Knowledge**

- Knowledge adds understanding, but without application it remains theoretical. The next step is action

- **Wisdom**

- Wisdom is the ability to apply knowledge judiciously to make sound decisions and take appropriate actions.

# Example

- **Data**

40, 55, 85, Computer Science

- **Information**

Marks obtained by a student in Computer Science

- **Knowledge**

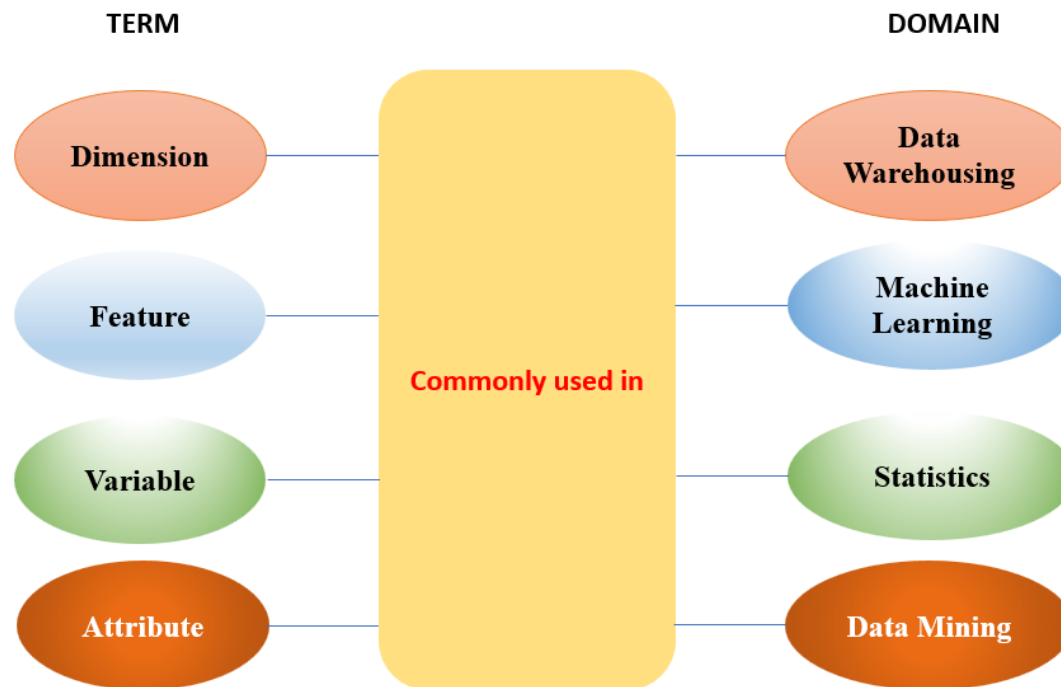
The student's performance is improving

- **Wisdom**

Encourage the student to pursue Computer Science

# Attributes

- An attribute is a data field, representing property or feature of a data object.
- Also known as **dimension**, **feature**, and **variable**.



# Categorical Attributes

- Categorical attributes, also known as qualitative or discrete attributes, represent data that falls into categories or groups rather than numerical values
- These categories can be represented by labels, symbols, or names.
- These attributes are further categorized into:
  - Nominal Attributes
  - Ordinal Attributes
- Categorical attributes play a vital role in data mining by providing qualitative information that drives pattern discovery, classification, segmentation, visualization, and interpretation of data, leading to informed decision-making and improved business outcomes.

# Nominal Attributes

- Nominal attributes represent categories with no inherent order or ranking. Each category is distinct and unrelated to the others.
- Examples of nominal attributes include:
  - Gender (Male, Female)
  - Marital Status (Single, Married, Divorced)
  - Eye Color (Blue, Brown, Green)
  - Country of Residence (USA, Canada, UK)
- In nominal attributes, there is no meaningful numerical comparison between categories.
- For instance, "Male" is not greater or lesser than "Female"; they are simply different categories.

# Ordinal Attributes

- Ordinal attributes, on the other hand, represent categories that have a clear order or ranking.
- While the categories are distinct, they also have a relative position or hierarchy.
- Examples of ordinal attributes include:
  - Education Level (High School, Bachelor's, Master's, Ph.D.)
  - Income Level (Low, Medium, High)
  - Customer Satisfaction Rating (Poor, Fair, Good, Excellent)
- In ordinal attributes, there is a meaningful order between categories, but the magnitude of differences between categories may not be uniform.
- For example, the difference between "Poor" and "Fair" in customer satisfaction may not be the same as the difference between "Fair" and "Good".



# Numerical Attributes

- Numeric attributes are represented by numbers and have most of the properties of numbers.
- These attributes can be either integer-valued or continuous.
- Numeric attributes are also called quantitative attributes.
- Numeric attributes can be interval-scaled or ratio-scaled.
- Understanding the types of numerical attributes—continuous and discrete—is crucial for effective data preprocessing, feature engineering, model selection, and interpretation in data mining tasks.

# Interval Scaled Attributes

- Interval scaled attributes represent numerical values where the intervals between consecutive values are equal and meaningful.
- These attributes have no true zero point, meaning that a value of zero does not indicate the absence of the attribute but rather represents a specific point on the scale.
- Examples of interval scaled attributes include temperature measured in Celsius or Fahrenheit, dates measured on the Gregorian calendar, and time measured in hours or minutes.
- Arithmetic operations such as addition and subtraction are meaningful for interval scaled attributes, but multiplication and division are not meaningful due to the lack of a true zero point.

# Ratio Scaled Attributes

- Ratio scaled attributes represent numerical values where both the intervals between consecutive values and the ratios of the values are meaningful.
- Examples of ratio scaled attributes include height measured in centimeters or inches, weight measured in kilograms or pounds, and counts of objects or events.
- Arithmetic operations such as addition, subtraction, multiplication, and division are all meaningful for ratio scaled attributes.

# Ratio Scaled Attributes

- The ratio scaled attributes have meaningful intervals and also a true zero point. True zero point means the absence of the property like 0 inches means there is no length.
- For example weight in KG and here 10 KG is twice as much as five KG (i.e.  $10 / 5$  is 2 or 2 times as much). So that's a meaningful ratio
- and in addition to this zero pounds means no weight or an absence of weight so there's a true zero point.

| Attribute Type               |          | Description                                                                                                                                                              | Examples                                                                                              |
|------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Categorical<br>(Qualitative) | Nominal  | The values of a nominal attribute are just different names; i.e., nominal values provide only enough information to distinguish one object from another.<br>(=, $\neq$ ) | zip codes,<br>employee ID numbers,<br>eye color, gender                                               |
|                              | Ordinal  | The values of an ordinal attribute provide enough information to order objects.<br>( $<$ , $>$ )                                                                         | hardness of minerals,<br>{ <i>good</i> , <i>better</i> , <i>best</i> },<br>grades,<br>street numbers  |
| Numeric<br>(Quantitative)    | Interval | For interval attributes, the differences between values are meaningful, i.e., a unit of measurement exists.<br>(+, -)                                                    | calendar dates,<br>temperature in Celsius<br>or Fahrenheit                                            |
|                              | Ratio    | For ratio variables, both differences and ratios are meaningful.<br>(*, /)                                                                                               | temperature in Kelvin,<br>monetary quantities,<br>counts, age, mass,<br>length,<br>electrical current |

# Comparison Chart

| Type     | Differences   | Ordering | True Zero | Ratios | Example             |
|----------|---------------|----------|-----------|--------|---------------------|
| Nominal  | ×             | ×        | ×         | ×      | Gender, Colors      |
| Ordinal  | √ (rank only) | √        | ×         | ×      | Survey rankings     |
| Interval | √             | √        | ×         | ×      | Temperature (°C)    |
| Ratio    | √             | √        | √         | √      | Height, Age, Income |

# Continuous Numerical Attributes

- Continuous numerical attributes represent values that can take any real number within a certain range.
- These values are often measured and can have an infinite number of possible values within the specified range.
- Examples of continuous numerical attributes include:
  - Temperature (measured in Celsius or Fahrenheit)
  - Height (measured in meters or feet)
  - Weight (measured in kilograms or pounds)
  - Time (measured in hours, minutes, seconds)
  - Income (measured in currency, e.g., dollars, euros)
- Continuous numerical attributes are characterized by their smooth and uninterrupted nature, and they can often be represented with decimal precision.

# Discrete Numerical Attributes

- Discrete numerical attributes represent values that are countable and finite, typically taking on integer values.
- These values are often obtained through counting or enumeration and do not have a continuous range.
- Examples of discrete numerical attributes include:
  - Number of children in a family
  - Number of products purchased in a transaction
  - Number of bedrooms in a house
  - Number of defects in a manufacturing process
  - Number of students in a classroom
- Discrete numerical attributes are characterized by their distinct and separate nature, where values are integers and cannot be subdivided further.



# Structured Data

- **Structured data** refers to data that is organized in a well-defined format with a fixed schema, typically arranged in rows and columns.
- Each data element belongs to a specific field with a predefined data type, which makes the data easy to store, retrieve, and process using database management systems.
- It is commonly stored in relational databases and spreadsheets, where it can be efficiently queried using languages such as SQL.
- Examples include student records, employee databases, and financial transactions.
- Because of its organized nature, structured data ensures consistency, accuracy, and fast access for analysis and reporting.

# Unstructured Data

- Unstructured data is data that does not have a predefined format or organization, making it difficult to store, search, or analyze using traditional databases.
- It can be in the form of text, images, audio, video, or social media posts, where the content does not fit neatly into rows and columns.
- Unstructured data lacks a consistent data model, so specialized tools and techniques like NoSQL databases, data mining, or machine learning are often needed to process it.
- Examples include videos, photographs, PDF documents, emails, and social media content.

# Semi Structured Data

- Semi-structured data is a type of data that does not conform to a strict table-like structure like structured data, but still contains organizational elements such as tags, labels, or markers that make it easier to interpret and process.
- It provides flexibility to store complex or variable information while allowing some level of analysis and querying.
- Semi-structured data is commonly found in formats like JSON, XML, CSV files, and emails with metadata, where the data may have varying fields or hierarchical structures.
- This type of data is widely used in web applications, data exchange, and logging systems because it balances flexibility and processability.

# Data Life Cycle

- Data Life Cycle refers to the complete series of stages that data undergoes during its existence, from creation to eventual deletion or archival.
- It encompasses how data is generated, stored, used, shared, maintained, and ultimately destroyed, ensuring its accuracy, security, and usefulness throughout its life.
- The cycle emphasizes accuracy, security, privacy, and compliance at each stage.
- It helps organizations manage data efficiently and responsibly.

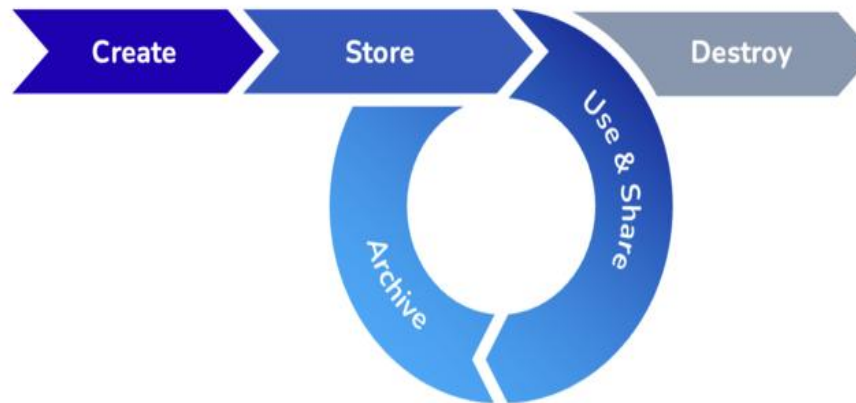


Fig. Phases of Data Life Cycle

# Creation and acquisition

- Data and information are created by people by manual input, or they arise in systems, applications, sensors by some measurement, generation, scanning, automatic recording, and so on.
- Individual systems and applications can be interconnected and so data arises between them by exchange, integration.
  - manual input and natural creation of data by people, users
  - natural creation by systems, applications, sensors, and so on
  - generation
  - integration, imports, and data exchange between systems and applications, receiving data from an external source

# Storing Information

- It is important to emphasize how closely the lifecycle is linked to the carrier, i.e., where and how data, information, or knowledge are stored.
- Data are acquired and recorded in some form into some system and on some specific medium.
- Digital or paper form can be destroyed, knowledge protection can only be contractually ensured, because it is difficult to destroy a person as their carrier in regular business practice.
  - digital (stored in an information system, software) or
  - printed in paper form on paper or
  - exist in someone's mind

# Using and Sharing

- Data exist to be used; unused data are meaningless.
- Information security is essential during data use.
- The life cycle of data is closely linked to its carrier (digital, paper, or human).
- Data can be recorded, stored, and destroyed; knowledge protection often requires contracts.
- **Key protections:**
  - Access control – restrict who can view or modify data
  - Shared data protection – safeguard distributed information

# Archiving

- Data Archiving is a stage in the data life cycle where data that is no longer actively used but still important for future reference or compliance is stored safely for the long term.
- This does not mean that they cease to exist, but they are subject to archiving (for example, because we need them to prove some fact).
- The main purpose is to:
  - Preserve data for regulatory compliance, audits, or historical reference.
  - Free up primary storage by moving inactive data to cheaper storage solutions.
- **Examples:**
  - Old financial records, employee files, research data, or historical transaction logs.
  - Archived emails or documents that are no longer actively edited.



# Destruction

- Data Destruction is the final stage of the data life cycle, where data that is no longer needed or has reached the end of its retention period is permanently deleted or destroyed.
- This ensures that sensitive information cannot be recovered or misused.
- The main purpose is to:
  - Protect privacy and confidentiality.
  - Comply with legal and regulatory requirements.
  - Free up storage space.
- It happens after data is no longer needed for operations or legal compliance or when archived data reaches the end of its retention period.

# Data Warehouse

- A **data warehouse** is a data repository that stores large volumes of structured data from multiple sources.
- According to W. H. Inmon (1992) who is a leading architect in construction of data warehouse systems, a data warehouse is a **subject-oriented, integrated, time-varying, non-volatile** collection of data in support of management's decision -making process.
- It is designed to support **business intelligence (BI)** activities, such as **reporting, data analysis, and decision making**.
- The main purpose of a data warehouse is to enable organizations to analyze historical data and generate meaningful insights to support strategic decisions.

# Characteristics of Datawarehouse

- A data warehouse has several defining characteristics that distinguish it from other types of databases.
- These are commonly known as the **Inmon's four key characteristics**:
- **Subject-Oriented:**
  - Data is organized around major subjects such as customers, sales, products, or finance.
  - Focuses on high-level entities rather than day-to-day operations.
- **Integrated:**
  - Combines data from various sources (e.g., databases, flat files, cloud platforms) into a consistent and unified format.
  - Ensures consistency in naming conventions, measurements, and encoding.

# Characteristics of Datawarehouse

- **Time Variant**

- Stores historical data, enabling trend analysis over time.
- Each record is associated with a time element (e.g., date of transaction, time period).
- The time horizon for the data warehouse is significantly longer than that of operational systems.
- Operational database: current value data. (60 to 90 days)
- Data warehouse data: provide information from a historical perspective (e.g., past 5-10 years)

- **Non-Volatile**

- Once data is entered into the warehouse, it is not updated or deleted.
- Ensures data stability and consistency for reporting and analysis.
- Data Warehouse is relatively static in nature.
- Data warehouse data is loaded and accessed, but it is not updated (in the general sense).

# Differences Between Operational Database and Data Warehouse

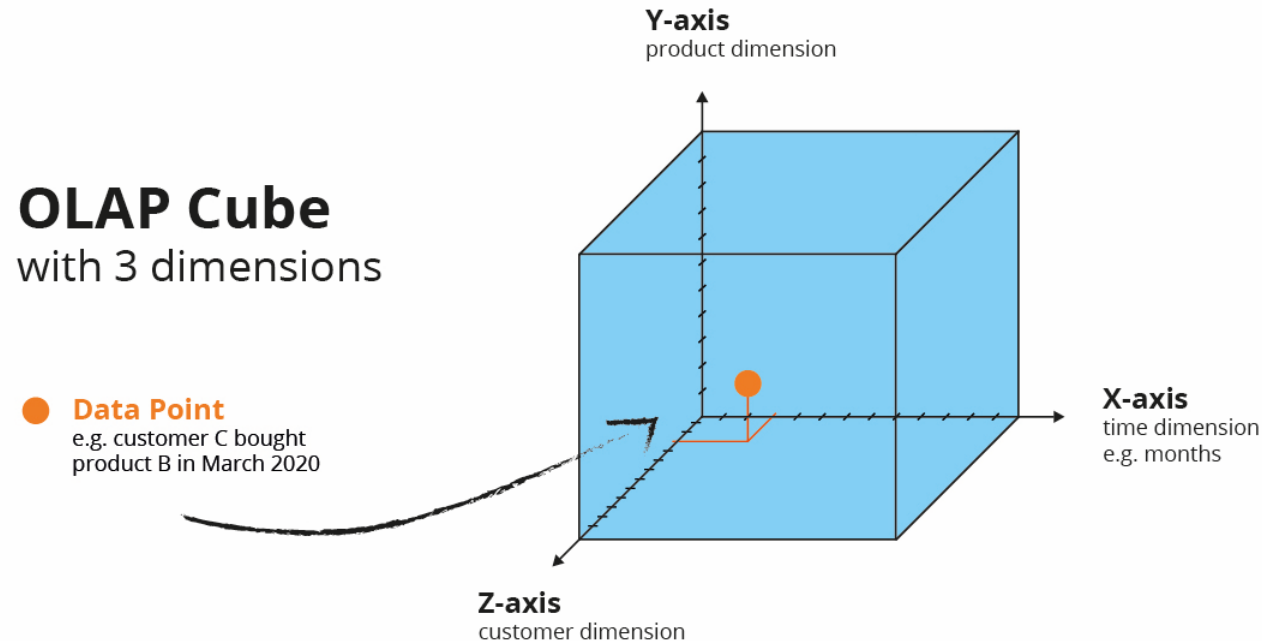
|                                       | <b>Data Warehouse</b>     | <b>Operational Database</b> |
|---------------------------------------|---------------------------|-----------------------------|
| <b>Purpose</b>                        | Analysis, Decision making | Day-to-Day                  |
| <b>Supports</b>                       | OLAP                      | OLTP                        |
| <b>Users</b>                          | Managerial community      | Clerical Community          |
| <b>Data Model</b>                     | Multi-dimensional         | Relational                  |
| <b>Age of Data</b>                    | Current and time series   | Current and real time       |
| <b>Data Modification</b>              | Read/Access only          | Insert, update and delete   |
| <b>Types of Data</b>                  | Static                    | Dynamic                     |
| <b>Amount of Data per Transaction</b> | Larger                    | Smaller                     |

# A Multi Dimensional Data Model

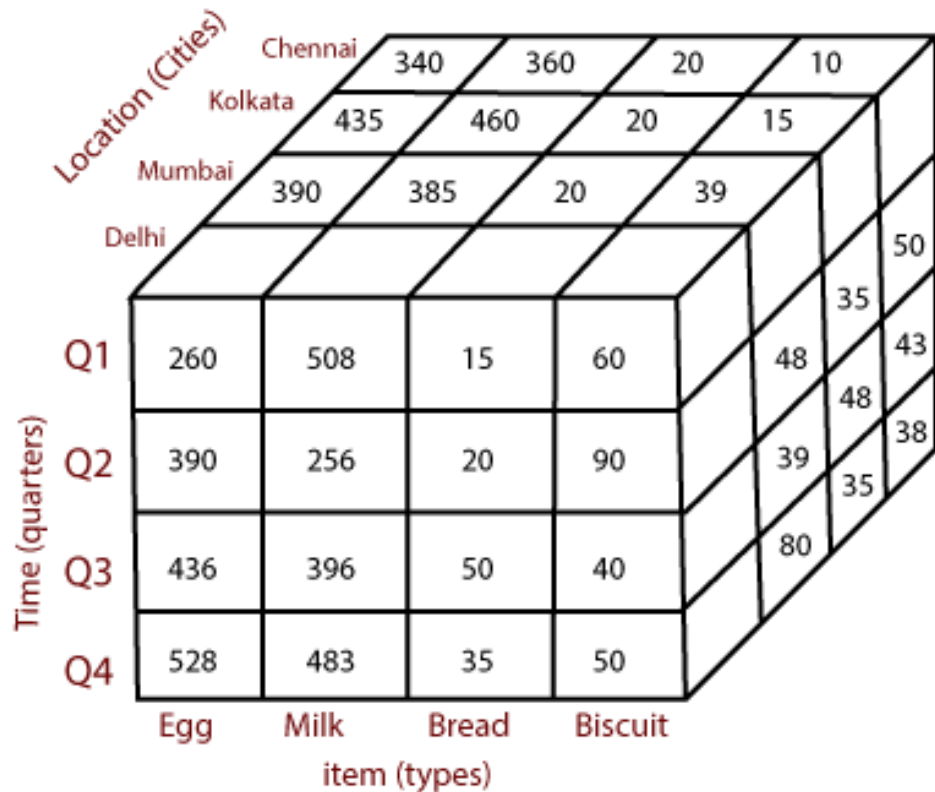
- A **dimension** is a perspective with respect to which we analyze the data
- A Datawarehouse is based on a **multidimensional data model** which views data in the form of a data cube
- A **multidimensional data model** is usually organized around a central theme (e.g., sales). Numerical measures on this theme are called **facts**, and they are used to analyze the relationships between the dimensions
- Example:
  - Central theme: sales
  - Dimensions: item, customer, time, location, supplier, etc.

# Data Cube

- The cube has one or more dimensions, such as time, geography, and product, and one or more measures, such as sales revenue, units sold, and profit margin.
- The **data cube** summarizes the measure with respect to a set of n dimensions and provides summarizations for all subsets of them



# Data Cube



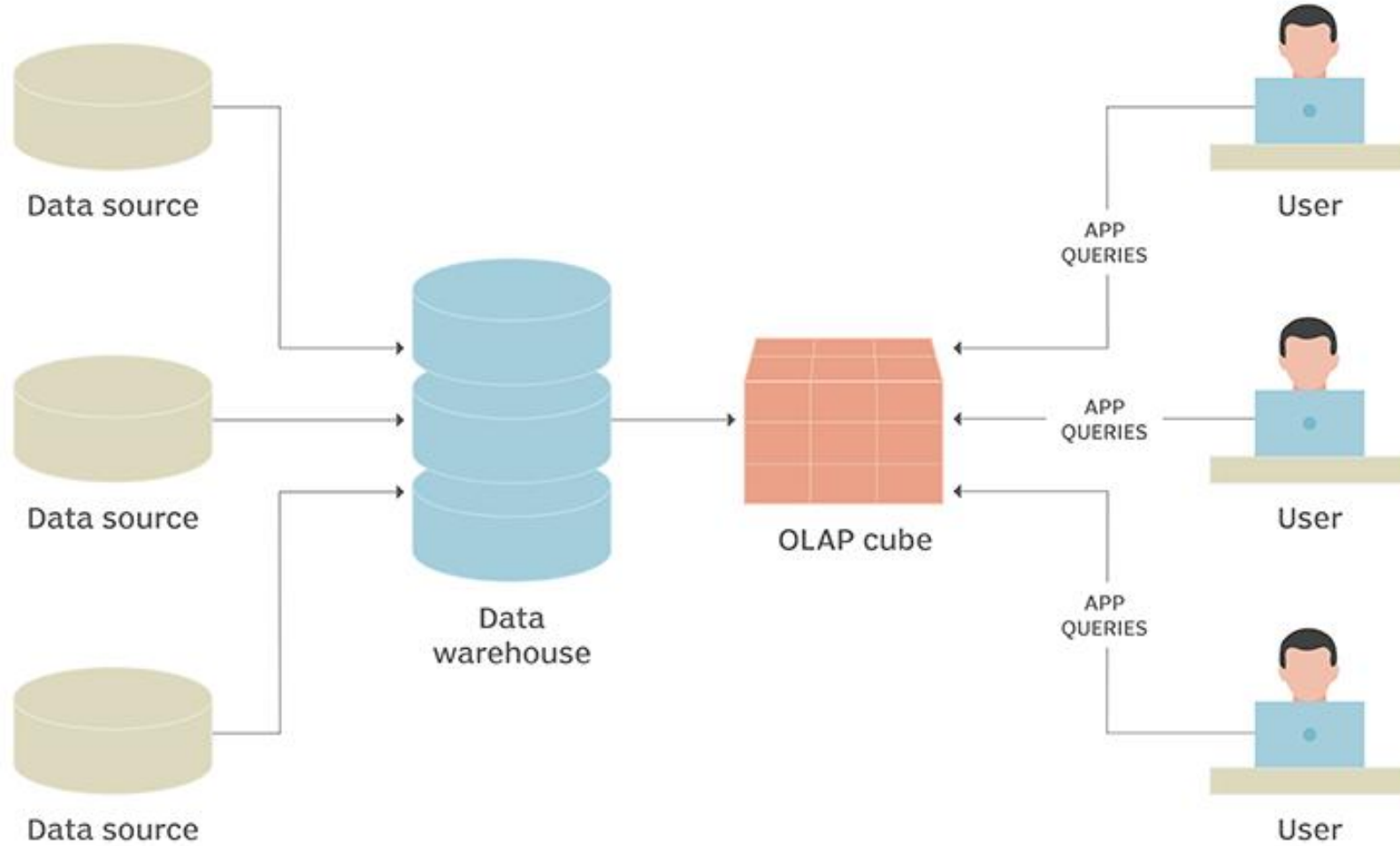
- Measures are the numerical values that are being analyzed, such as sales revenue, units sold, or profit margin.
- Each measure is associated with one or more dimensions.
- The data cube is created by aggregating and summarizing data along multiple dimensions, resulting in a multi-dimensional array or matrix of data.



# Online Analytical Processing

- OLAP (Online Analytical Processing) is a technique for analyzing large amounts of data from multiple perspectives or dimensions.
- The data is organized into a multidimensional data structure, such as a cube or a hypercube.
- OLAP allows users to quickly and easily explore and analyze data to identify trends, patterns, and outliers, and make informed decisions.

# OLAP Process



# Characteristics of OLAP Systems (FASMI)

- **Fast:** Delivers information to the user at a fairly constant rate i.e.  $O(1)$  time. Most queries answered in under 5 seconds.
- **Analysis:** Performs basic numerical and statistical analysis of the data, pre-defined by an application developer or defined ad-hocly by the user.
- **Shared:** Implements the security requirements necessary for sharing potentially confidential data across a large user population.
- **Multi-dimensional:** OLAP data is organized into multiple dimensions, such as time, geography, and product, which allows users to analyze data from different perspectives.
- **Information:** Accesses all the data and information necessary and relevant for the application, wherever it may reside and not limited by volume.

# Why OLAP?

- **Analyze sales performance:** OLAP allows businesses to analyze sales data from different dimensions, such as product, geography, and time, to identify which products are selling well in which regions, and to track sales performance over time.
- **Understand customer behavior:** OLAP enables businesses to analyze customer data, such as purchase history, demographics, and preferences, to gain insights into customer behavior and preferences, and to develop targeted marketing campaigns.
- **Optimize operations:** OLAP can help businesses to optimize operations by analyzing data related to inventory, production, and supply chain, and identifying areas for improvement and cost savings.
- **Improve decision-making:** By providing users with fast and easy access to relevant data, OLAP can help businesses to make informed decisions, such as whether to launch a new product, enter a new market, or invest in a particular area of the business.

# Example

- Let's say you're a business analyst at a retail company and you want to analyze sales data to understand which products are selling well in different regions. You have a large data set with information about products, customers, regions, and sales.
- With OLAP, you can create a multidimensional data cube that allows you to analyze sales data from different dimensions. For example, you could create a cube with the following dimensions:  
**Product, Region and Time.**
  - The Product dimension might have categories such as clothing, electronics, and home goods.
  - The Region dimension might have categories such as North America, Europe, and Asia.
  - The Time dimension might have categories such as year, quarter, and month.
- You could then query the data cube to answer questions like:
  1. Which products are selling the most in North America in Q1 of 2022?
  2. What is the total revenue for clothing sales in Europe in 2021?
  3. Which regions are showing the most growth in sales of electronics products over the past year?
- By using OLAP to analyze the data, you can easily slice and dice the data across different dimensions and quickly get answers to these types of questions.
- This can help you identify trends, opportunities, and areas for improvement in your business.

# OLAP vs OLTP

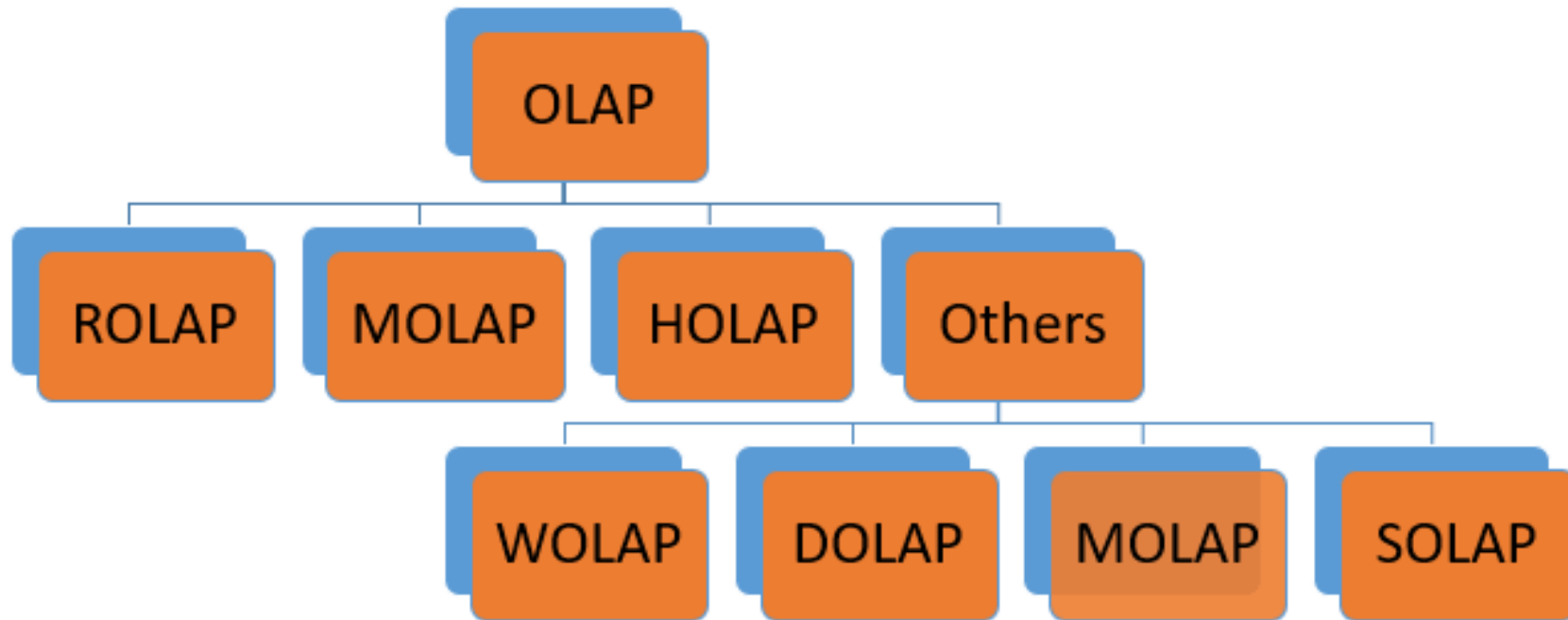
- OLTP is designed for transactional processing, while OLAP is designed for analytical processing. OLTP is optimized for high transactional performance, while OLAP is optimized for high analytical performance
- **Purpose:** OLTP systems are designed to support transactional processing, which involves inserting, updating, and deleting records in a database in real-time. OLAP systems, on the other hand, are designed to support analytical processing, which involves querying and analyzing data to gain insights and make informed decisions.
- **Data Structure:** OLTP systems typically use a normalized data model, which means that data is stored in a highly structured and normalized form to support efficient transaction processing. OLAP systems, on the other hand, typically use a denormalized data model, which means that data is stored in a way that supports efficient query processing and multidimensional analysis.
- **Data Volume:** OLTP systems typically handle high volumes of transactional data, which is processed in real-time. OLAP systems, on the other hand, typically handle large volumes of historical data, which is analyzed and queried in a batch or near-real-time manner.
- **Performance:** OLTP systems are optimized for high transactional performance, which means that they are designed to support many concurrent users and high transaction volumes. OLAP systems are optimized for high analytical performance, which means that they are designed to support complex queries and multidimensional analysis.
- **User Type:** OLTP systems are designed for transactional users, such as sales clerks, customer service representatives, and order processors, who need to quickly access and update data in real-time. OLAP systems are designed for analytical users, such as business analysts, data scientists, and executives, who need to analyze and understand business data to make informed decisions.

# OLTP vs. OLAP

|                           | <b>OLTP</b>                                                  | <b>OLAP</b>                                                             |
|---------------------------|--------------------------------------------------------------|-------------------------------------------------------------------------|
| <b>users</b>              | clerk, IT professional                                       | manager                                                                 |
| <b>function</b>           | day to day operations                                        | Decision support                                                        |
| <b>DB design</b>          | application-oriented                                         | subject-oriented                                                        |
| <b>data</b>               | current, up-to-date<br>detailed, flat relational<br>isolated | historical,<br>summarized, multidimensional<br>integrated, consolidated |
| <b>usage</b>              | repetitive                                                   | ad-hoc                                                                  |
| <b>access</b>             | read/write<br>index/hash on prim. key                        | lots of scans                                                           |
| <b>unit of work</b>       | short, simple transaction                                    | complex query                                                           |
| <b># records accessed</b> | tens                                                         | millions                                                                |
| <b>#users</b>             | thousands                                                    | hundreds                                                                |
| <b>DB size</b>            | 100MB-GB                                                     | 100GB-TB (even PB)                                                      |
| <b>metric</b>             | transaction throughput                                       | query throughput, response                                              |

# Types of OLAP

- **OLAP Hierarchical Structure**





# Types of OLAP

- **Relational OLAP(ROLAP):**

ROLAP is an extended RDBMS along with multidimensional data mapping to perform the standard relational operation.

- **Multidimensional OLAP (MOLAP)**

MOLAP Implements operation in multidimensional data.

- **Hybrid OnlineAnalytical Processing (HOLAP)**

In HOLAP approach the aggregated totals are stored in a multidimensional database while the detailed data is stored in the relational database. This offers both data efficiency of the ROLAP model and the performance of the MOLAP model.

# Types of OLAP

## **Desktop OLAP (DOLAP)**

In Desktop OLAP, a user downloads a part of the data from the database locally, or on their desktop and analyze it.  
DOLAP is relatively cheaper to deploy as it offers very few functionalities compares to other OLAP systems.

## **Web OLAP (WOLAP)**

Web OLAP which is OLAP system accessible via the web browser. WOLAP is a three-tiered architecture. It consists of three components: client, middleware, and a database server.

## **Mobile OLAP:**

Mobile OLAP helps users to access and analyze OLAP data using their mobile devices

## **Spatial OLAP :**

SOLAP is created to facilitate management of both spatial and non-spatial data in a Geographic Information system (GIS)

# ROLAP

ROLAP works with data that exist in a relational database. Facts and dimension tables are stored as relational tables. It also allows multidimensional analysis of data and is the fastest growing OLAP.

## Advantages of ROLAP model:

- **High data efficiency.** It offers high data efficiency because query performance and access language are optimized particularly for the multidimensional data analysis.
- **Scalability.** This type of OLAP system offers scalability for managing large volumes of data, and even when the data is steadily increasing.

## Drawbacks of ROLAP model:

- **Demand for higher resources:** ROLAP needs high utilization of manpower, software, and hardware resources.
- **Aggregately data limitations.** ROLAP tools use SQL for all calculation of aggregate data. However, there are no set limits to the for handling computations.
- **Slow query performance.** Query performance in this model is slow when compared with MOLAP

# MOLAP

- MOLAP uses array-based multidimensional storage engines for multidimensional views of data.
- MOLAP is more optimized for fast query performance and retrieval of summarized information.

## Advantages of MOLAP

- Excellent performance.
- Can perform complex calculations.

## Dis-Advantages of MOLAP

- Limited amount of data handled only.
- Require additional investment.

# Hybrid OLAP (HOLAP)

Hybrid OLAP is a mixture of both ROLAP and MOLAP. It offers fast computation of MOLAP and higher scalability of ROLAP. HOLAP uses two databases.

1. Aggregated or computed data is stored in a multidimensional OLAP cube
2. Detailed information is stored in a relational database.

## Benefits of Hybrid OLAP:

- This kind of OLAP helps to economize the disk space, and it also remains compact which helps to avoid issues related to access speed and convenience.
- Hybrid HOLAP's uses cube technology which allows faster performance for all types of data.
- ROLAP are instantly updated and HOLAP users have access to this real-time instantly updated data. MOLAP brings cleaning and conversion of data thereby improving data relevance. This brings best of both worlds.

## Drawbacks of Hybrid OLAP:

- Greater complexity level: The major drawback in HOLAP systems is that it supports both ROLAP and MOLAP tools and applications. Thus, it is very complicated.
- Potential overlaps: There are higher chances of overlapping especially into their functionalities.

# OLAP Operations in Multi Dimensional Data Model

- Data cube operations are operations performed on data cubes to extract information from them. Some common data cube operations include:
- **Roll-up**
  - This operation aggregates data from a lower level of granularity to a higher level of granularity.
  - For example, rolling up a sales data cube from daily sales to monthly sales.
- **Drill-down**
  - This operation breaks down data from a higher level of granularity to a lower level of granularity.
  - For example, drilling down a sales data cube from monthly sales to daily sales.
- **Slice**
  - This operation selects a subset of the data cube based on a specific value or range of values for one or more dimensions.
  - For example, selecting data from a sales data cube for a specific region.

# Data Cube Operations

## Dice

- This operation selects a subset of the data cube based on a specific combination of values for two or more dimensions.
- For example, selecting data from a sales data cube for a specific region and time period.

## Pivot

- This operation rotates the data cube to view it from a different perspective.
- For example, pivoting a sales data cube to view sales by product instead of by region.

Locations  
(countries)

USA

Canada

2000

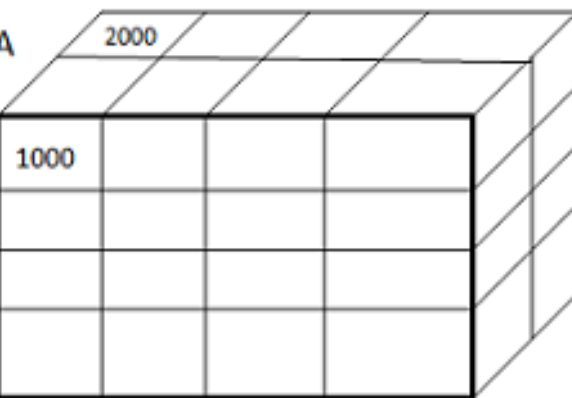
Time  
(Quarter)

Q1

Q2

Q3

Q4



Mobile Modem Phone Security  
item(types)

**roll-up** on location  
(from cities to countries)

Locations  
(cities)

Chicago

New York

Toronto

Vancouver

440

1560

395

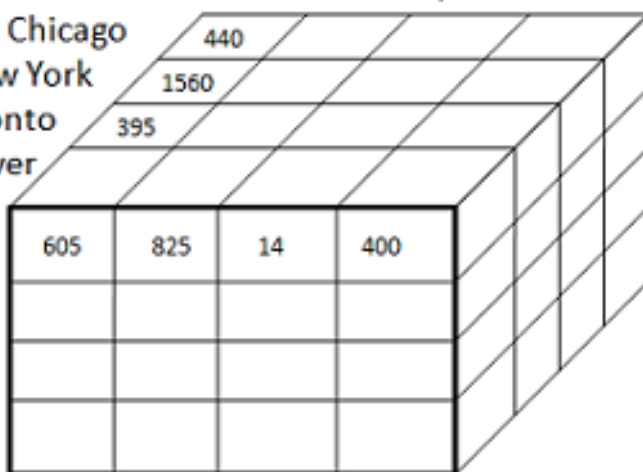
Time  
(Quarter)

Q1

Q2

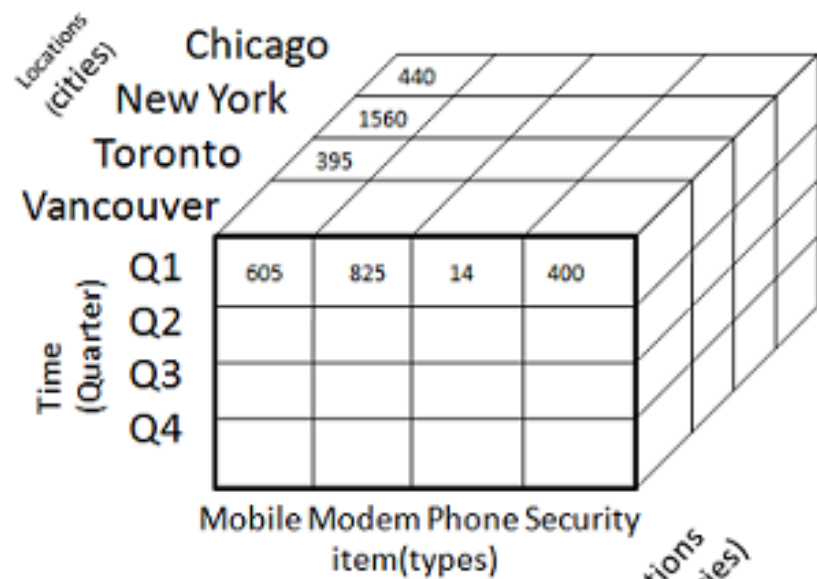
Q3

Q4

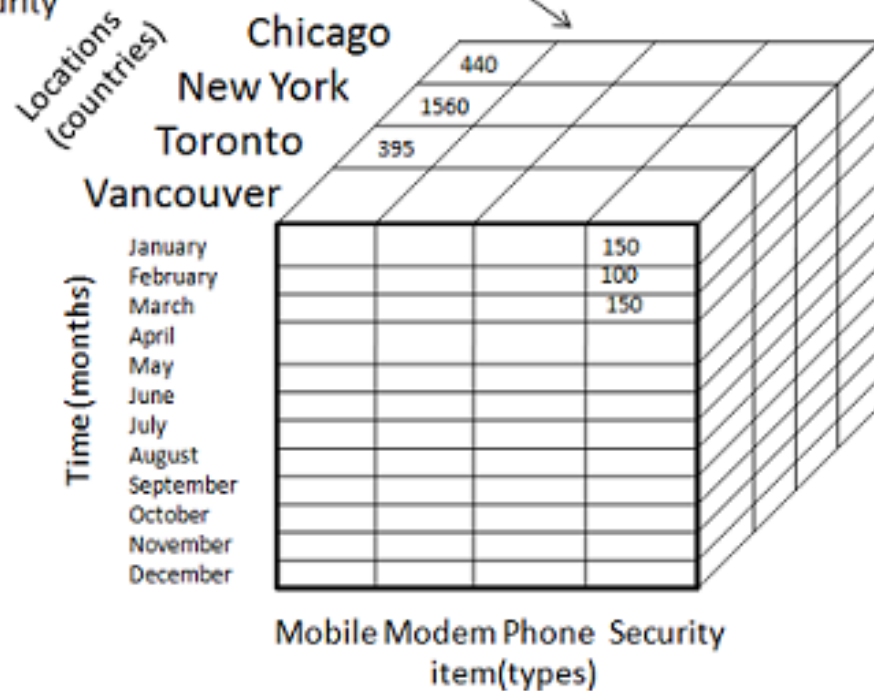


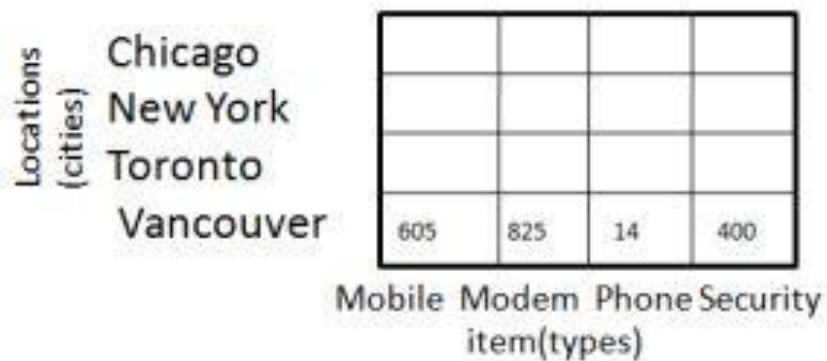
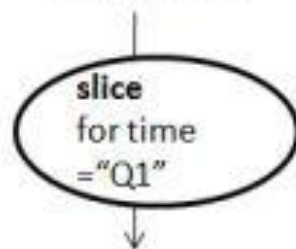
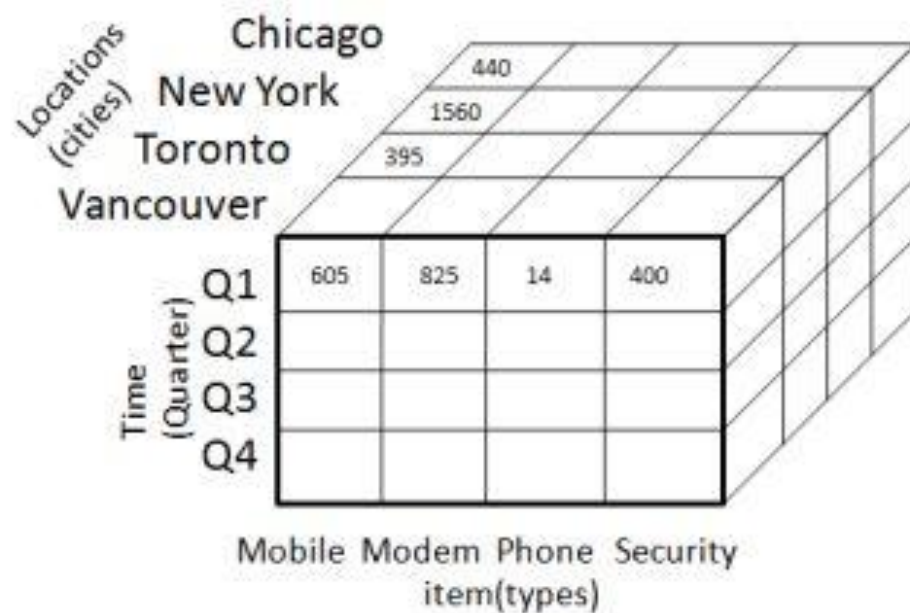
Mobile Modem Phone Security  
item(types)

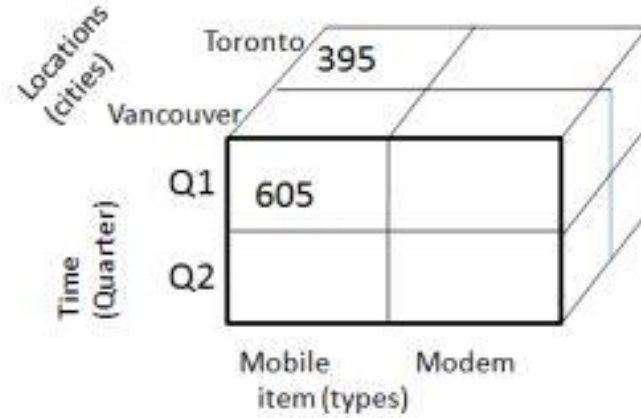




Drill down on  
time(from  
quarters to month)



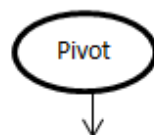




Dice for (location = "Toronto" or "Vancouver")  
and (time = "Q1" or "Q2") and  
(item = "Mobile" or "Modem")



|                       |           |             |       |       |          |
|-----------------------|-----------|-------------|-------|-------|----------|
| Locations<br>(cities) | Chicago   |             |       |       |          |
|                       | New York  |             |       |       |          |
|                       | Toronto   |             |       |       |          |
|                       | Vancouver | 605         | 825   | 14    | 400      |
|                       |           | Mobile      | Modem | Phone | Security |
|                       |           | item(types) |       |       |          |



|                 |          |                   |          |         |           |
|-----------------|----------|-------------------|----------|---------|-----------|
| Item<br>(types) | Mobile   |                   |          |         | 605       |
|                 | Modem    |                   |          |         | 825       |
|                 | Phone    |                   |          |         | 14        |
|                 | Security |                   |          |         | 400       |
|                 |          | Chicago           | New York | Toronto | Vancouver |
|                 |          | Location (Cities) |          |         |           |

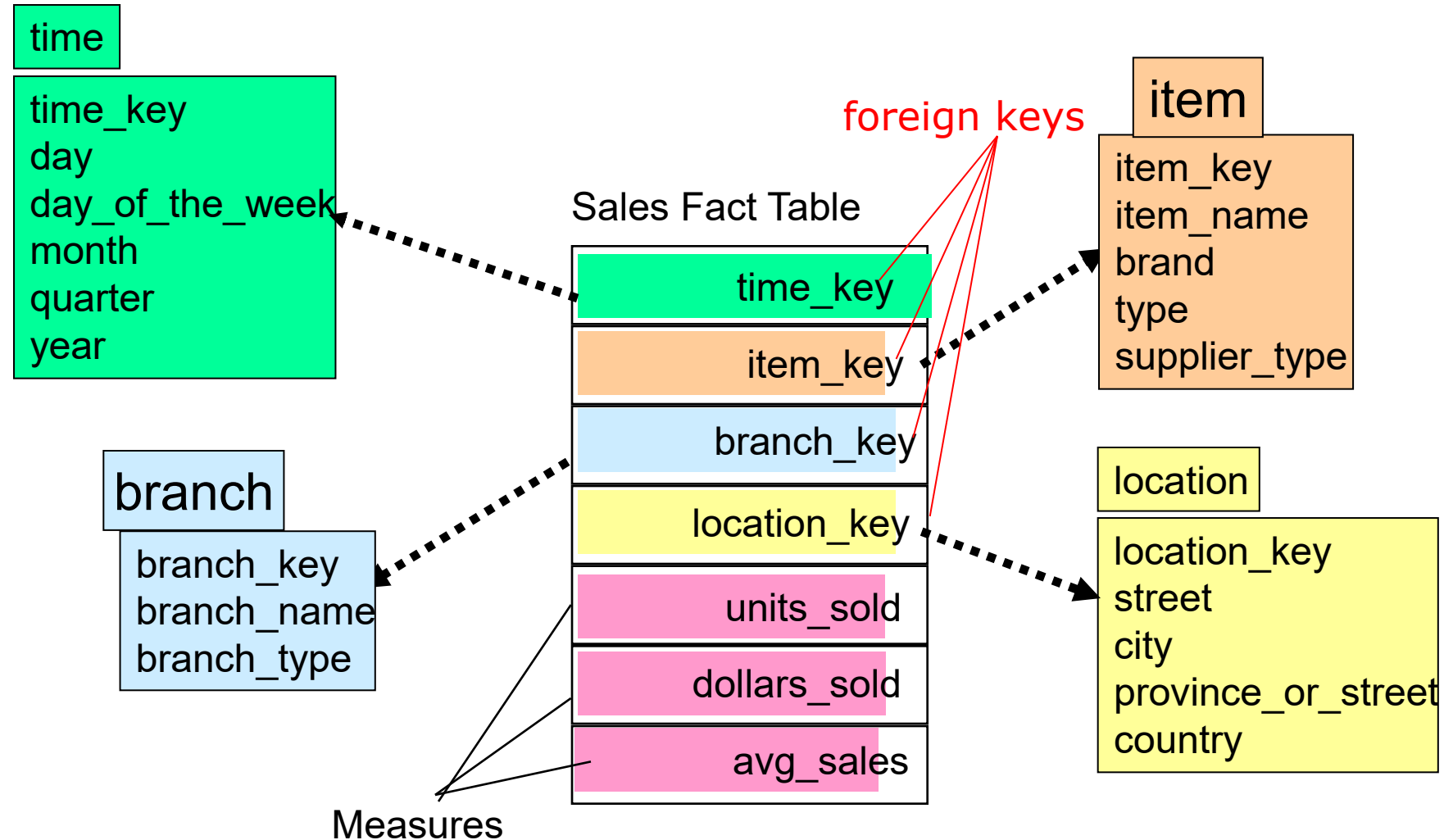
# Conceptual Modeling of Data Warehouse

- The conceptual modeling develops a schema for logical representation of data stored in data warehouse.
- A data warehouse requires a concise, subject-oriented schema that facilitates on-line data analysis.
- The most popular data model for a data warehouse is a multidimensional model.
- Such a model can exist in the form of *a star schema*, *a snowflake schema*, or *a fact constellation schema*.
  - **Star schema:** A fact table in the middle connected to a set of dimension tables
  - **Snowflake schema:** A refinement of star schema where some dimensional hierarchy is normalized into a set of smaller dimension tables, forming a shape similar to snowflake
  - **Fact constellations:** Multiple fact tables share dimension tables, viewed as a collection of stars, therefore called galaxy schema or fact constellation

# Star Schema

- In a star schema, the data warehouse is organized around a central fact table, which contains the measures or numerical data that the organization wants to analyze.
- For example, a fact table might contain sales revenue, number of units sold, or customer count.
- Surrounding the fact table are one or more dimension tables, which provide context to the measures in the fact table.
- Dimension tables contain descriptive data about the business, such as customer demographics, product categories, or time periods.
- Each dimension table is joined to the fact table by a foreign key relationship.
- The star schema gets its name from the shape of the diagram that represents the relationship between the fact table and the dimension tables, which looks like a star.

# Example of Star Schema



# Star Schema

- **Advantages:**

- Easy to understand since all the information about each level is stored in one row.
- A star schema optimizes performance by keeping queries simple and providing fast response time as only one join requires to establish the relationship between the fact table and any one of the dimension table.

- **Disadvantages:**

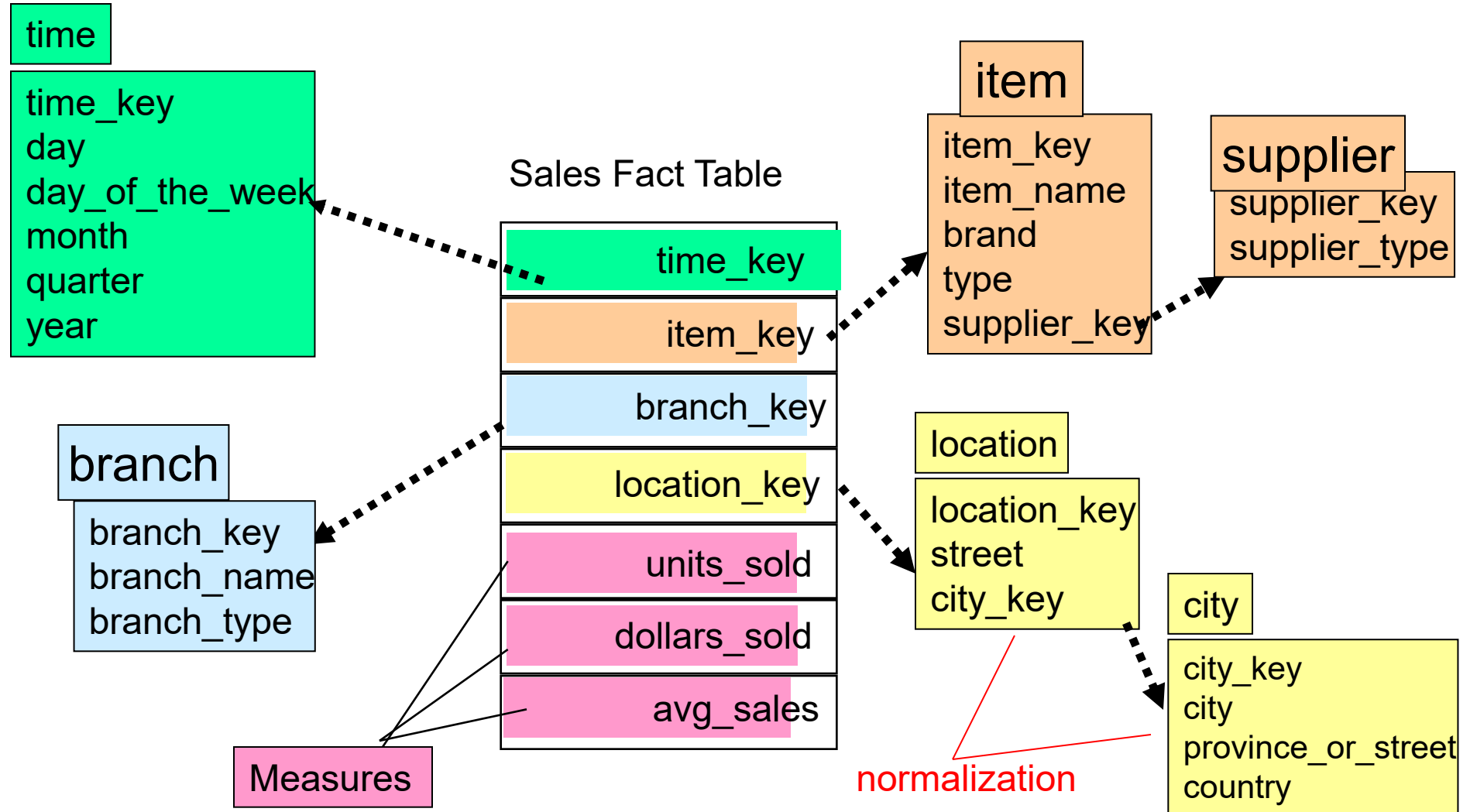
- Redundancy of the data hence occupies additional space



# Snowflake Schema

- The snowflake schema is a variant of the star schema model, where some dimension tables are *normalized*, thereby further splitting the data into additional tables.
- Snowflake schemas normalize dimensions to eliminate redundancy.
- While this saves space, it increases the number of dimension tables and requires more foreign key joins.
- The result is more complex queries and reduced query performance.

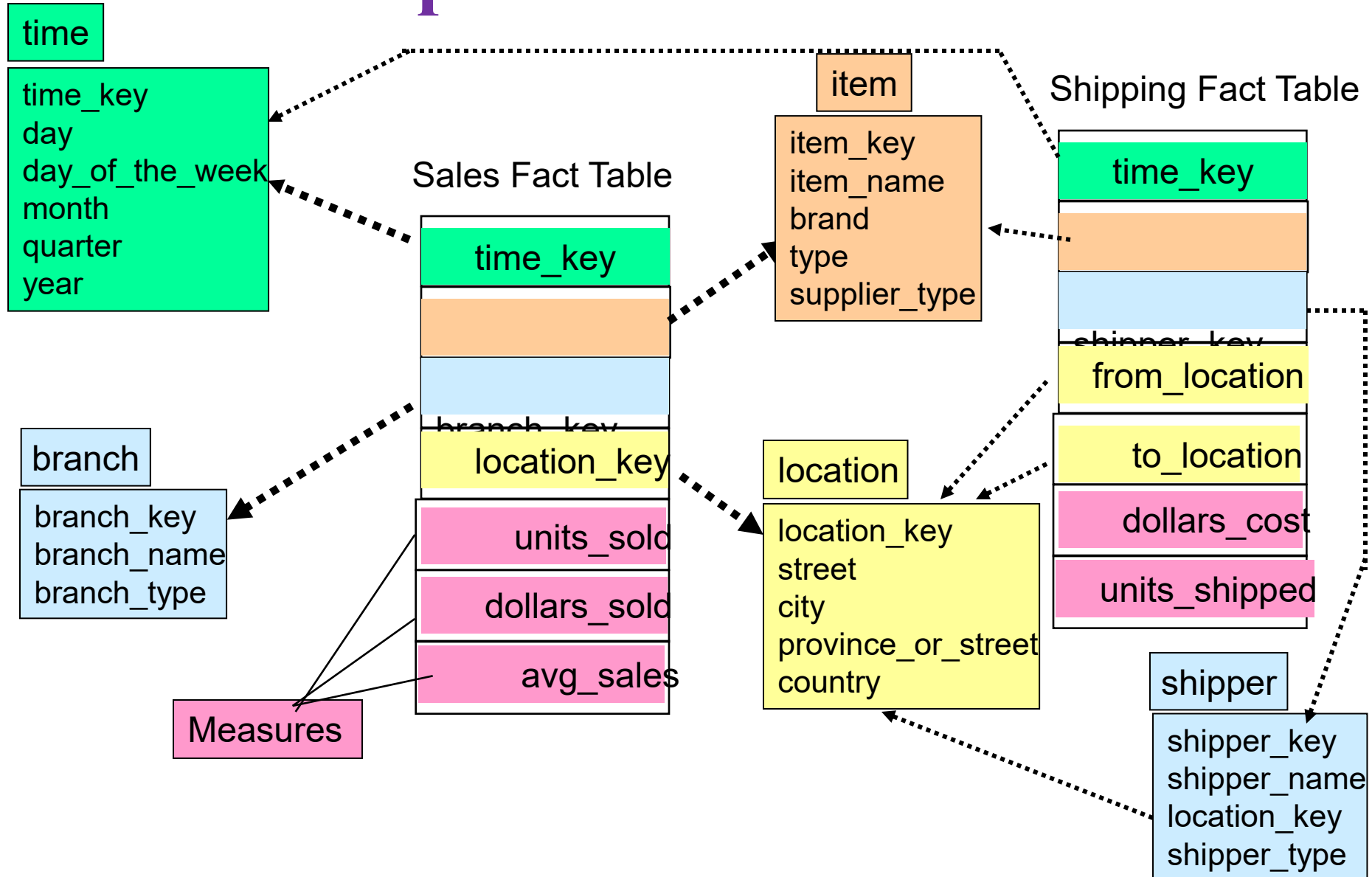
# Example of Snowflake Schema



# Fact Constellation schema

- A Fact constellation schema is a type of schema which consists of more than one fact table sharing to dimension tables.
- This kind of schema can be viewed as a collection of stars, and hence is called a galaxy schema or a fact constellation.

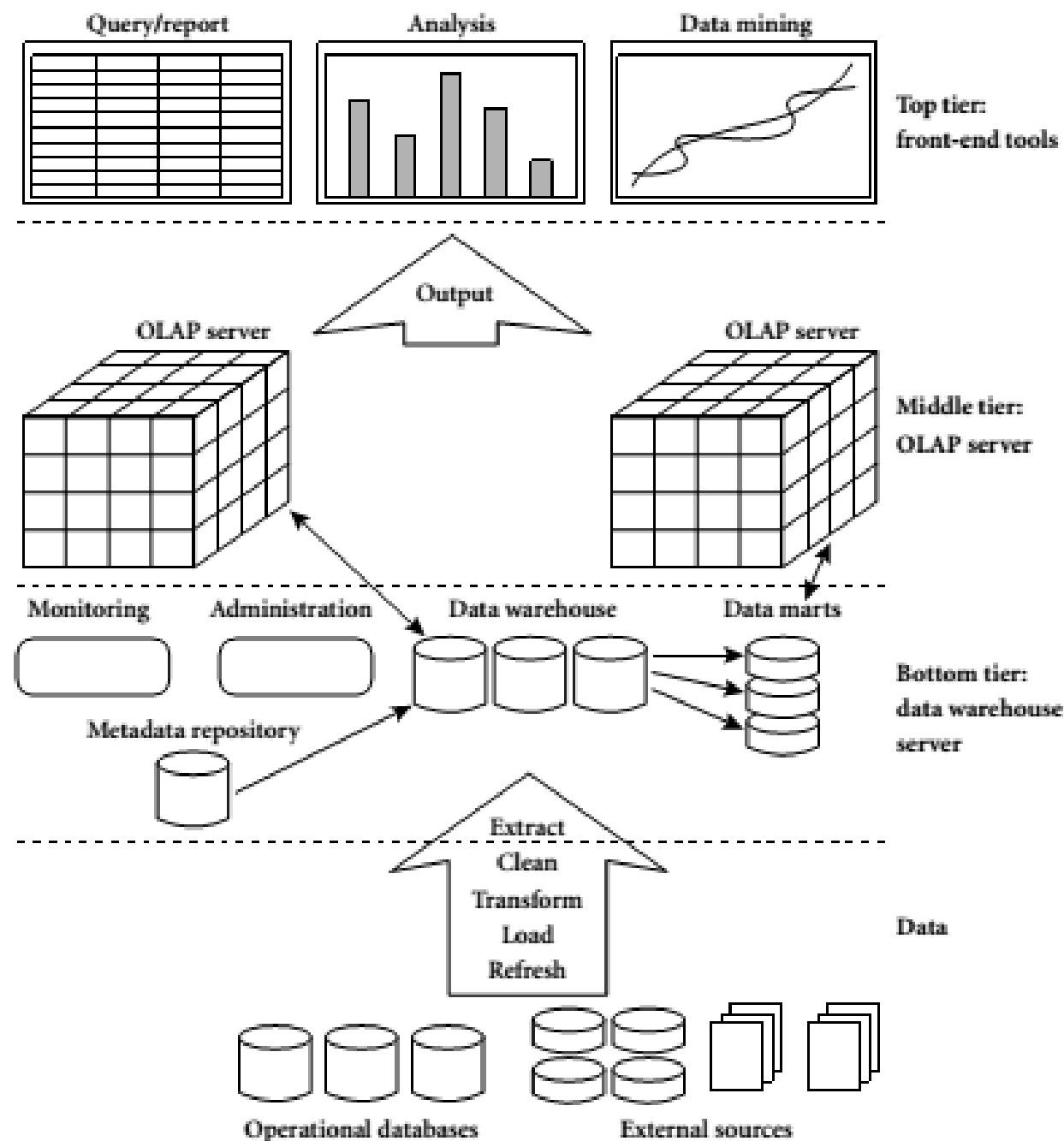
# Example of Fact Constellation



# Data Warehouse Architecture

- Data warehouse architecture refers to the design and structure that defines how data is collected, stored, processed, and accessed in a data warehouse system.
- A three-tier data warehouse architecture is the most widely used architecture in real-world data warehouse systems.

Figure:  
A three-tier  
Data  
warehousing  
Architecture



# Data Warehouse Architecture

1. **Data Source:** A data warehouse system uses heterogeneous sources of data either from operational databases or from some external sources (such as customer profile information provided by external consultants).
2. **Data warehouse server:** Data from heterogeneous sources are stored to one logically centralized single repository: a data warehouse through the extraction, cleaning, transformation, load and refresh functions. Data warehouse can be directly accessed, but it can be also used as a source for creating data marts. Metadata repositories stores information on sources, access procedures, data staging, data mart schema and so on.

# Data Warehouse Architecture

3. **OLAP server:** OLAP server that is typically implemented using either a relational OLAP (ROLAP) model or a multidimensional OLAP (MOLAP) model.
4. **Front-end client tools:** It contains query and reporting tools, analysis tools, and/or data mining tools.



# Data Warehouse Implementation

- Data warehouse implementation refers to the process of designing, building and deploying a centralized system that stores, manages and integrates data from various sources to support business analysis and decision-making.
- The purpose of implementing a data warehouse is to provide organizations with a unified repository of historical data, enabling efficient querying, reporting and data analysis.
- Proper implementation is crucial for businesses because it ensures the system meets the organization's needs for scalability, data accuracy and performance.
- A well-implemented data warehouse provides key benefits such as improved decision-making, streamlined data access, enhanced reporting capabilities and better data consistency, all of which contribute to data-driven business success.

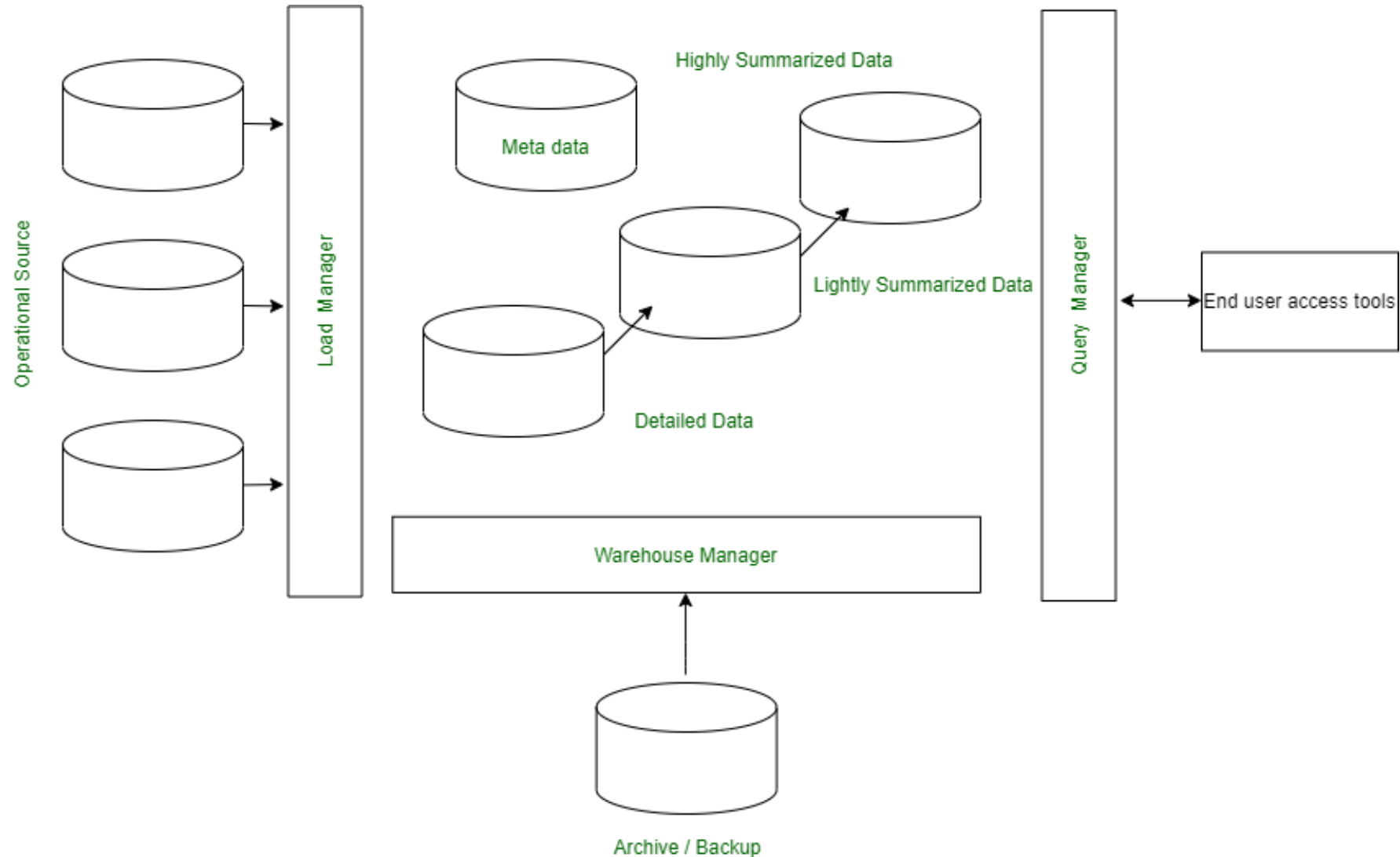
# Steps in Data Warehouse Implementation

- **Planning and Requirements Gathering:** Understand business needs, set objectives and determine hardware/software requirements.
- **Data Modeling and Design:** Choose the appropriate schema (e.g., Star, Snowflake) to optimize data organization and query performance.
- **ETL Process: Extract, Transform and Load:** Extract, clean, transform and load data, ensuring consistency and accuracy, with customized ETL(extract, transform and load) tools.
- **Database Design and Architecture:** Design the physical architecture, including storage, indexing and optimization for efficient performance.
- **Data Warehouse Development:** Create tables, views and other objects, ensuring scalability for future data growth.
- **Testing and Validation:** Verify data accuracy and performance, ensuring the system meets requirements and resolving any discrepancies.
- **Deployment and Maintenance:** Deploy the system, address issues, apply updates and maintain continuous data integration.

# Components of data warehouse

- A data warehouse is an integrated system that consolidates data from operational systems and external sources, providing valuable insights for decision-making.
- It includes various components that work together to store, manage and access data efficiently.
- The data moves from the data source area through the staging area to the presentation server. The entire process is better known as ETL.

# Components of data warehouse



# Components of data warehouse

## 1. Operational Source Systems

- Provide raw data from internal systems (e.g., relational databases like Informix oracle) and external sources.
- Serve as the primary input for the data warehouse.

## 2. Load Manager

- Manages the ETL (Extract, Transform, Load) process for data extraction and transformation.
- Prepares data for entry into the warehouse, ensuring it meets the required format.

## 3. Warehouse Manager

- Oversees data storage, aggregation and analysis within the data warehouse.
- Handles tasks like data de-normalization, backup, collection and optimization for better performance.

## 4. Query Manager

- Handles user queries within the data warehouse.
- Supports querying, reporting and data retrieval, with functionality dependent on the available end-user tools.

# Components of data warehouse

## 5. Detailed Data

- Stores granular, raw data for complex analysis and reporting.
- Provides comprehensive insights and supports in-depth research.

## 6. Summarized Data

- Stores predefined aggregations of detailed data for faster queries and reports.
- Provides high-level insights to assist in decision-making.

## 7. Archive and Backup Data

- Ensures data integrity and disaster recovery through regular backups and archival storage.
- Archives older, less frequently accessed data for future use or compliance.

## 8. Metadata

- Metadata contains information about data structure, source and transformation processes.
- Supports the ETL process, warehouse management and querying by providing essential context for data.

## 9. End-User Access Tools

- Serve as the interface for users to interact with the data warehouse.
- Include analysis, reporting and data mining tools, enabling users to access, query and derive insights from the data.

# Data Warehouse Models

- Enterprise Warehouse
- Data Marts
- Virtual Warehouse

# Enterprise warehouse

- An enterprise warehouse collects all of the information about subjects spanning the entire organization.
- It provides corporate-wide data integration, usually from one or more operational systems or external information providers.
- An enterprise data warehouse may be implemented on traditional mainframes, computer superservers, or parallel architecture platforms.
- It requires extensive business modeling and may take years to design and build.



# Data Marts

- Data mart is a subset of data warehouse built specifically for department.
- The scope is confined to specific selected subjects.
- For example, a marketing data mart may confine its subjects to customer, item, and sales.
- ***Independent data marts*** are sourced from data captured from one or more operational systems or external information providers, or from data generated locally within a particular department or geographic area.
- ***Dependent data marts*** are sourced directly from enterprise data warehouses.

# Virtual warehouse

- A virtual warehouse is a set of views over operational databases.
- For efficient query processing, only some of the possible summary views may be materialized.
- A virtual warehouse is easy to build but requires excess capacity on operational database servers.

# Assignment #1

## Question:

With the rapid growth of data in organizations, traditional databases are no longer sufficient for analytical purposes. Justify this statement by explaining the need for Data Warehousing and discuss the current trends in Data Warehousing that address these challenges.

**Thank You !**